

Transit Ridership Analysis — Summer 2025 Bus Network

A strategic, C-level analysis of **Summer 2025 APC stop-summary data** covering **127 bus and trackless trolley routes**, **18,088 stop-direction records**, and **4,293,991 weekly riders**. The workbook answers four strategic executive questions — Revenue & Profitability, Operational Efficiency, Strategic Growth & Network Design, and Resource Allocation — through dashboards, charts, and drill-down tables.

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Project Goal

Analyze APC stop-level bus activity to identify high-volume routes and stops, service-day patterns, directional imbalances, revenue/cost performance, and network-design opportunities — translating raw stop-summary data into executive-ready answers to four strategic questions.

Key Findings at a Glance

1. Revenue & Profitability

- **64 routes (50%) operate below break-even** on the cost proxy (\$2.25 fare per boarding; cost = \$50/stop + \$0.10 per weekly rider).
- **Route 47 is the star performer** with **167,195 weekly riders** — the clear #1 route in the network.
- Only 20 of 127 routes (Premier + High tiers) average above ~80K weekly riders; the bottom 26 routes form the underperforming tail.

2. Operational Efficiency

- **Routes with <200m stop spacing perform best** (~37K average weekly ridership) — tighter spacing correlates with higher demand capture.
- **Pareto concentration: the top 10% of stops carry 55.4% of all ridership.** Average stop spacing network-wide is 199m, with an average of 43 boardings per stop.

3. Strategic Growth & Network Design

- **Frankford Transit Center is the #1 hub** — 12 routes, 36,797 combined weekly riders — followed by the 69th St Transit Center complex (three stops combining for ~95K weekly riders).
- **39 routes are commuter-heavy** (<15% weekend usage); only 1 route is weekend-heavy (>30%), confirming a weekday-commuter-dominated network.

4. Resource Allocation

- **34.5% of ridership occurs within 5km of the city center**, and ridership per stop decays sharply with distance (543 avg riders/stop within 0–2km vs. 38 beyond 30km).
- **Routes 47 and 23 show massive directional activity gaps** between weekday and weekend flows, suggesting schedule-optimization opportunities by direction and day type.

Key Metrics

Metric	Value
Total weekly ridership (all routes)	4,293,991
Routes analyzed	127
Stops analyzed	18,088
Avg weekly ridership per route	33,811
Top performing route	Route 47 (167,195 weekly)
Routes below break-even (proxy)	64
Weekend revenue as % of total	18.6%
Avg boardings per stop	43
Top 10% stops carry % of ridership	55.4%
Major transfer hubs (2+ routes)	20
Commuter-heavy routes (<15% weekend)	39
Weekend-heavy routes (>30%)	1
Avg stop spacing	199m
Ridership within 5km of center	34.5%

Dataset Overview

- **Source sheet:** Summer_2025_StopSummary_(Bus)
- **Grain:** one row per route × direction × stop × stop code
- **Records:** 18,088 data rows (+1 header row), columns A:Q
- **Routes:** 127 | **Directions:** Eastbound, Westbound, Northbound, Southbound, Loop
- **Source system:** APC (Automated Passenger Counter)
- **Core ridership fields:** Weekdays_0, Weekdays_1, Saturdays_, Saturdays1, Sundays_0n, Sundays_0f
- **Geography:** latitude/longitude retained as numeric fields for map-ready analysis

Workbook Guide

Sheet	Contents
4_Strategic_Exec_Questions	The four strategic question categories with key findings

Table 2 – continued

Sheet	Contents
Executive Summary	C-level overview: key questions, metrics, and workbook contents
Route Rankings	Top 15 routes by weekly ridership, with tier classification
Day Type Breakdown	Weekday vs. Saturday vs. Sunday ridership by route
Performance Matrix	Scatter chart: Stars vs. Workhorses vs. Question Marks vs. Dogs
Tier Distribution	Pie chart: Premier / High / Medium / Low route tiers
Top Busiest Stops	Top 20 individual stops + hourly/detail context
Directional Imbalance	Net flow (boardings – alightings) analysis + bar chart
Revenue vs Cost	Scatter chart with break-even line; ROI by route
Distance Analysis	Ridership density decay by distance band from city center
Transfer Hubs	Multi-route stops ranked by combined weekly ridership
Route Direction Performance	Top/bottom route-direction combinations by day type
Summer_2025_StopSummary_(Bus)	Raw APC source data (preserved as source of truth)
Pivot_Table_1	Supporting pivot aggregation
Readme Documentation	In-workbook documentation of cleaning and build steps

Route Tier Distribution

Tier	Routes	% of Network	Avg Weekly Ridership
Premier	9	7.1%	122,637
High	11	8.7%	79,758
Medium	33	26.0%	45,947
Low	74	58.3%	10,766

Methodology

1. Data Cleaning

- 1. Preserved the raw worksheet as the source of truth for audit and rework.
- 2. Verified dataset boundaries: columns A:Q, rows 1:18,089.
- 3. Validated headers and identified key fields: route, direction, stop code, stop name, latitude, longitude, and day-type ons/off.
- 4. Standardized field interpretation: `_0/0n`-style columns = boardings; `_1/0ff`-style columns = alightings.

5. Treated route values as labels (not pure numbers) so mixed route identifiers (e.g., B1 OWL, BLVDDIR) stay stable in summaries and charts.
6. Confirmed ridership columns are numeric and aggregation-ready.
7. Retained zero-ridership rows — they can represent valid directional stops, terminals, or stops with no observed activity on a given service day.
8. Left stop names unaltered so every summary traces back to the original stop label.

2. Feature Engineering

1. Service-day totals combining ons and offs per day type.
2. Total boardings (weekday + Saturday + Sunday ons) and total alightings (offs).
3. Total activity = all ons + offs across all day types.
4. Day-of-week activity shares for weekday/weekend comparison.
5. Route-level, direction-level, and stop-level aggregations for rankings and imbalance detection.
6. Cost proxy: **\$2.25 fare per boarding; cost = \$50/stop + \$0.10 per weekly rider** for break-even and ROI analysis.

3. Dashboard Build Process

1. Built summary KPIs first so dashboards answer key questions before showing detail.
2. Route-ranking and stop-ranking views to surface top performers.
3. Day-type and direction comparisons for demand-pattern evaluation.
4. Charts for high-level comparison (11 embedded charts); tables for drill-down.
5. Consistent number formatting (thousands separators) and standalone titles/labels so screenshots work directly in this README.
6. Verified all summaries use the full 18,088 records — no sampling.

Validation Numbers

Measure	Value
Weekday boardings	350,708
Weekday alightings	348,452
Saturday boardings	229,764
Saturday alightings	228,381
Sunday boardings	170,482
Sunday alightings	169,564
Top route by total activity	Route 47 (59,183 combined daily ons/offs)

Assumptions & Notes

- Weekdays_1, Saturdays1, and Sundays_0f are treated as alighting fields based on their pairing with boarding fields.
- Revenue/cost figures use the proxy assumptions above and are directional, not accounting-grade.
- Break-even classification depends on the cost proxy; sensitivity to fare and per-stop cost assumptions should be considered before operational decisions.
- Weekday figures in the source are per-day averages; weekly totals use weekday × 5.

Suggested Screenshots

1. **Executive Summary** — KPI cards and key metrics overview.
2. **Performance Matrix** — Stars vs. Dogs quadrant scatter chart.
3. **Revenue vs Cost** — scatter with break-even line (64 routes below).
4. **Tier Distribution** — Premier/High/Medium/Low pie chart.
5. **Day Type Breakdown** — weekday vs. weekend ridership comparison.
6. **Transfer Hubs** — top multi-route stops table/map.
7. **Distance Analysis** — ridership decay from city center.
8. **Directional Imbalance** — net flow bar chart.

How to Use This Workbook

1. Start at Executive Summary for the headline story.
2. Use 4_Strategic_Exec_Questions to navigate by business question.
3. Drill into the analysis sheets (Rankings → Matrix → Revenue vs Cost → Hubs → Distance) for supporting evidence.
4. Reference Summer_2025_StopSummary_(Bus) for any stop-level audit of the source records.